

Motion-Aware Image Interaction Using Temporal Additive Color Mixing and High-Speed Projection

Shio Miyafuji*
Institute of Science Tokyo
Tokyo, Japan
miyafuji@c.titech.ac.jp

Arisa Kohtani*
Institute of Science Tokyo
Tokyo, Japan
kohtani.a.aa@m.titech.ac.jp

Hideki Koike
Institute of Science Tokyo
Tokyo, Japan
koike@c.titech.ac.jp

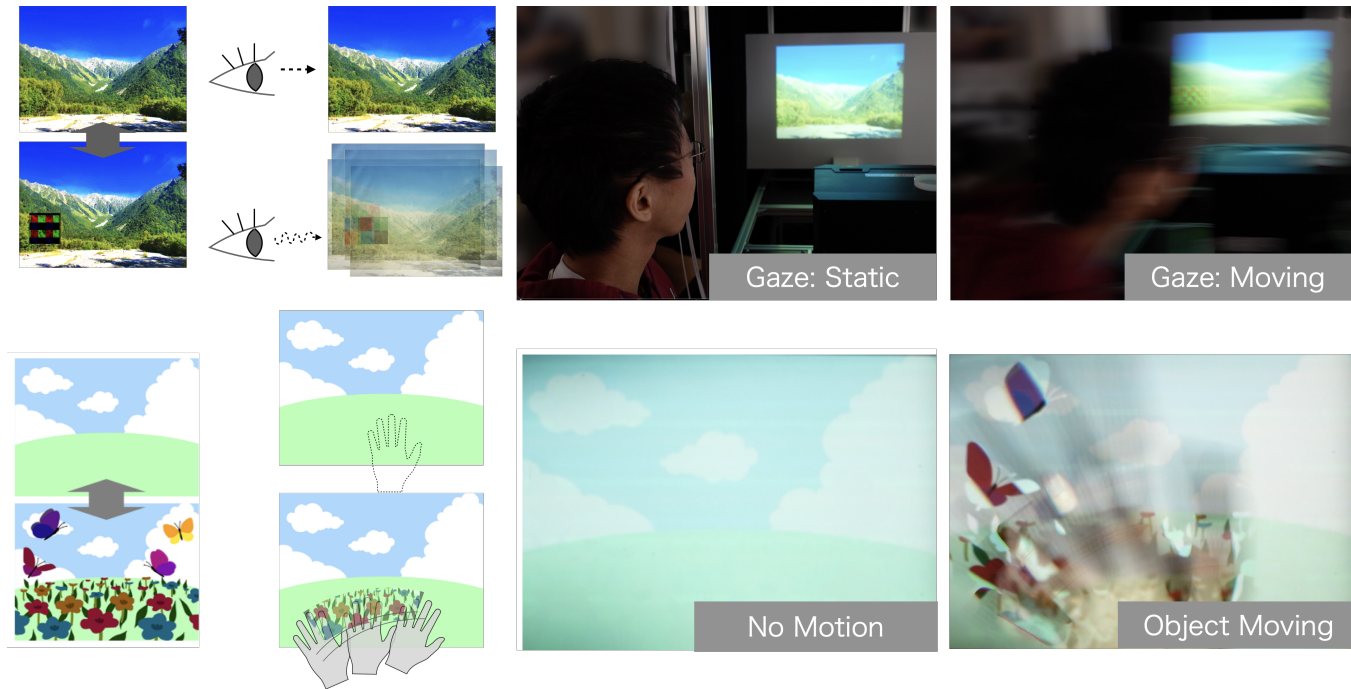


Figure 1: Motion-Aware Image Interaction Using Temporal Additive Color Mixing and High-Speed Projections: (Top) Users can observe gaze-guiding with mosaic patterns when their gazes are moving. (Bottom) Users can observe target images appearing in afterimages on moving objects.

Abstract

This study explores a novel interaction method in Spatial Augmented Reality (SAR) that enables direct interaction with images without relying on external devices. In the proposed method, a single image is decomposed into multiple frames and projected at high speed. Through temporal additive color mixing and afterimage effects, the image perceived by the user changes based on their gaze movement. This paper introduces two demonstrations based on different decomposition methods: Demo 1 uses mosaic patterns to

guide user gaze by displaying highly salient images during movement while allowing the original image to appear when stationary. Demo 2 adjusts the display ratios of the target and complementary images to make the target image visible in the afterimage of a moving object. The proposed method facilitates interaction solely through user or object movement, without requiring cameras or sensors, and holds potential for applications in public advertisements and signage.

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721182>

CCS Concepts

• Hardware → Emerging interfaces; • Human-centered computing → Displays and imagers; Mixed / augmented reality.

Keywords

High Speed Projection, Spatial Augmented Reality, Motion-aware Interaction, Afterimage Effect

ACM Reference Format:

Shio Miyafuji, Arisa Kohtani, and Hideki Koike. 2025. Motion-Aware Image Interaction Using Temporal Additive Color Mixing and High-Speed Projection. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3706599.3721182>

1 Introduction

The Spatial Augmented Reality (SAR) technology [4], which augments real-world space using projection, enables the display of desired images to multiple users without the need for additional devices such as head-mounted displays or any handheld displays. This capability suggests that various expression methods utilizing projection are expected to be applied in public spaces, such as for advertisements and guide signage. Interaction with publicly displayed images is generally achieved through methods such as extracting information from the images and presenting additional details on external devices. In particular, displaying QR codes in advertising is common. As a more intuitive approach, there are methods that embed information into images or light without altering the appearance of the original displayed image [1, 2]. These methods rely on terminal devices to read embedded information and present additional details on the device.

Additionally, methods for embedding information without device-based decoding or reading have also been proposed. Some studies have used the afterimage effect to present two-dimensional images using rapidly flashing LED lights [3, 11] and to alter perceived images according to the state of motion [5, 10]. While these methods utilize the afterimage effect to provide different perceptions depending on motion, they impose certain constraints on the speed or direction of the motion. Moreover, interaction techniques that do not rely on tracking or directional constraints have also been proposed. Isogawa et al. [6] proposed a method that enables the selection and manipulation of displayed information using shadows. By utilizing the additive properties of light, this approach enables information to be displayed exclusively within shadowed regions without requiring tracking of moving objects. While it eliminates motion direction and tracking constraints, it requires a complex multi-projector system. In contrast, Yamamoto et al. [12] proposed a method that uses high-frame-rate LED panels to reveal hidden text or images—ordinarily invisible—through hand-waving gestures. This method is simpler, requiring only a high-frame-rate LED panel. However, it is restricted to binary images for display. Building upon these previous works, we have proposed a method [7, 8] that overcomes their limitations by:

- minimizing constraints on motion direction,
- eliminating the need for tracking,
- operating without external devices,
- avoiding complex systems, and
- enabling the inclusion of full-color image.

In this demonstration, we show demos for novel interaction methods that enable direct interaction with images themselves without the use of external devices, utilizing high-speed projection and temporal additive color mixing. This technique uses high-speed projection to alternately display the target image and its complementary image at a speed that combines them into the original

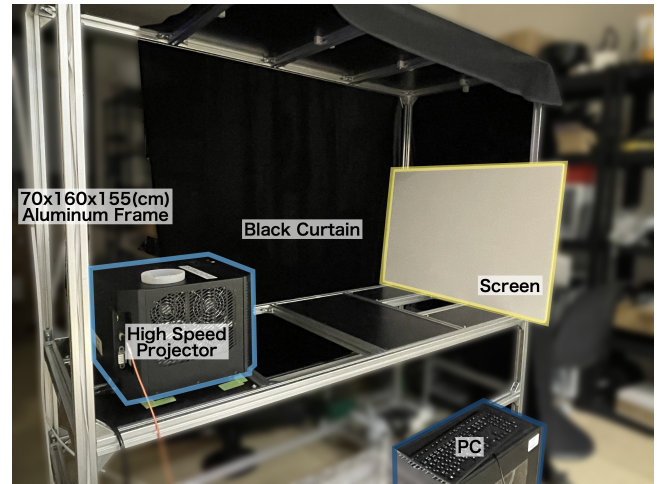


Figure 2: Demonstration setup: all hardwares are installed within a 70cm(width)×155cm(depth)×160cm(hight) aluminum frame.

image. By doing so, it controls the color-breaking phenomenon, allowing the target image to be observed when the user's gaze moves [8] or as an afterimage on objects [7].

2 Temporal Additive Color Mixing using High-Speed Projector

2.1 Hardware Setup

Figure 2 shows the setup used in the demonstration. The hardware used includes a high-speed projector (DynaFlash¹), a PC (CPU: Intel Core i7, GPU: RTX 3080 Ti) for image creation and projection control, and a white screen. These hardware components are installed within a 70cm(width)×155cm(depth)×160cm(hight) aluminum frame. The aluminum frame is covered with black curtains to ensure an optimal projection environment. The projected images can be observed by users by viewing the screen from the area around the back of the projector.

2.2 Basic Method: Temporal Additive Color Mixing

Conventional DLP projectors express colors by rapidly and sequentially displaying red, blue, and green light. This technique, which involves displaying multiple colors in quick succession to be perceived as a blended color by the human eye, is known as 'temporal additive color mixing'. However, when obstructions pass through the projection path or frame drops occur due to light source movement, the mixing process may fail, resulting in unintended colors being perceived. This phenomenon is called as 'color breaking' [13].

In our study, we aim to control the color-breaking phenomenon to create a system where nothing is visible when stationary, but the target image becomes visible when the user's gaze moves or objects within the user's field of view are in motion. In this method, the colors of an image are temporally decomposed into several

¹<https://us.teldevice.com/product/tb-6v-dynaflash/>

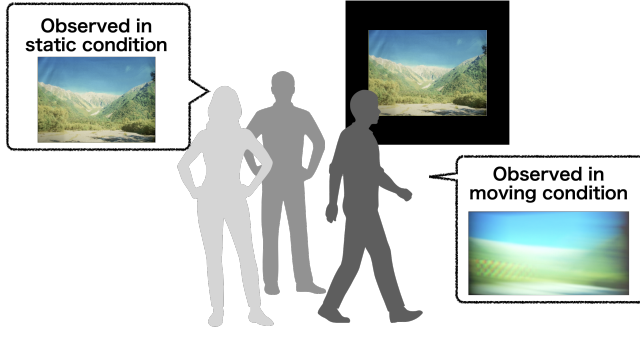


Figure 3: Demonstration overview of gaze guidance with the mosaic pattern decomposition.

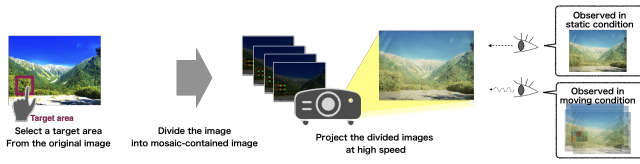


Figure 4: Overview of gaze guidance with the mosaic pattern decomposition.

image frames, which are then projected at high speed. As a result, color breaking caused by the decomposed image frames is observed by the human visual system only when the gaze or objects are in motion. In such case, however, if the display speed of each frame is too slow, flickering becomes noticeable to the human eye. The Critical Flicker Fusion Frequency (CFFF) in flickering two frames is about 50-90 Hz [9]. Based on this value, the actual projection speed is determined considering the surrounding environment and image brightness, ensuring that the speed between the decomposed frames are within the range of 60 Hz to 120 Hz. For example, when displaying an image decomposed into 4 frames, the projection speed is determined at 480 Hz. In the case of an image decomposed into 16 frames, the speed is at 960 Hz.

3 Demonstration

In this demonstration, we introduce two types of interaction methods between images and human vision by altering the image decomposition method: In Demo1, decomposition using mosaic patterns is applied to enhance the saliency within the image, enabling gaze guidance that responds to user movement. In Demo2, the image is decomposed into the target color and its complementary color. The proportions of the displayed colors are then adjusted, enabling specific colors to appear in the afterimages in response to the movement of objects.

3.1 Demo1: Gaze Guidance Using Decomposition with Mosaic Patterns

Demo 1 is Mosion [8], where the saliency within the image changes based on the user's gaze movement. In this demonstration, when the user's gaze is stationary, the original image is visible. Conversely, when the user's gaze moves, such as during walking, only specific

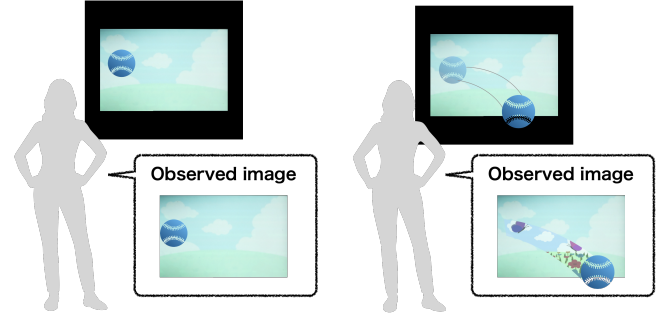


Figure 5: Demonstration overview of displaying the image in afterimage on the moving objects.

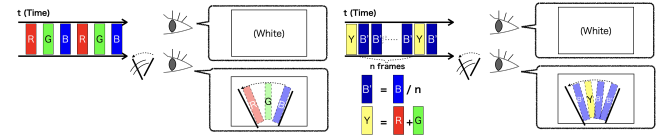


Figure 6: Overview of displaying in Afterimages by adjustment of color decomposition ratios. (Left) When the colors are evenly decomposed into R/G/B. (Right) When the colors are decomposed into the target color and its complementary color, and the target color is further divided into n frames.

areas with higher saliency become visible. Figure 4 illustrates an overview of the method. In this method, the areas intended to draw attention are decomposed into frames consisting of mosaics of red, green, blue, and black rectangles, which are then projected continuously at a high speed of 480 Hz. Figure 3 shows an overview of this demonstration. When the user's gaze moves, color breaking occurs in the areas decomposed into mosaic patterns. This results in the observer perceiving highly salient mosaics with intense color changes and multiple edges. Figure 1 (Top) shows a captured images during actual observation of the high-speed projection. Users who perceive the mosaic are guided to direct their gaze toward the mosaic area. However, when their gaze is stationary, the original image is recognized without the mosaic. This method, therefore, serves as a gaze-guiding technique that does not alter the appearance of the original image, which is a promising approach for applications such as public advertisements and road lighting.

3.2 Demo2: Image Display in Afterimages by Adjustment of Color Decomposition Ratios

Figure 5 shows an overview of this demonstration. Demo 2 is a demonstration where an image is displayed on the afterimage of an object as the user moves it within the projection space [7].

In this method, the target image is decomposed into frames consisting of the original image and a complementary image created using the complementary color for each pixel. Figure 5 illustrates an overview of this method. When the original image is a white image, and the frames are evenly decomposed into Red, Green, and Blue images (R/G/B), the afterimages appearing along the trajectory of a moving object will show R, G, and B, as Figure 5 (Left) shows.

To increase the proportion of the target color appearing in the afterimage, as Figure 6 (Right) shows, the target color in the image is further decomposed into n frames. These frames are then displayed in rapid succession, consisting of n frames of the target image and one frame of the complementary color of the image. This approach enables users to perceive the target color as if it is consistently present. Figure 1 (Bottom) shows the result of high-speed projection at 960 Hz, where 16 frames of the target image and 1 frame of the complementary image were displayed with $n = 16$. As shown in this figure, users can observe the target image appearing in the afterimage of the object's movement trajectory.

4 Conclusion

This demonstration introduced an interaction method with images mediated by human visual characteristics, utilizing a high-speed projector and temporal additive color mixing. This method decomposes the colors used in the original image using a specific technique and projects them at high speed, causing the appearance of the image to change based on user or object movement. Using this approach, we introduced two demonstrations showcasing novel interactions with images. In Demo 1, we introduced a gaze-guiding demonstration where the original image is decomposed into images containing a rectangular mosaic pattern. This allows moving users to see a highly salient image that guides their gaze, while stationary users can perceive the original image. In Demo 2, we introduced a demonstration where the original image is decomposed into the target image and its complementary image. By adjusting the display time ratio of each image, the target image is made visible in the afterimage of a moving object within the projection space. These methods have certain limitations, such as the inability to decompose colors that are not included in the original image and the need to account for variations in how the target image appears based on individual visual characteristics. Nonetheless, the capability to interact with images solely through user movement, without requiring additional cameras, sensors, or devices as long as the projection is displayed, highlights its potential. Therefore, This method is expected to enable a wide range of applications, such as public advertisements and signage.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Number JP20H04221 and JST, PRESTO Grant Number JPMJPR23I9, Japan.

References

- [1] Satoshi Abe, Atsuro Arami, Takefumi Hiraki, Shogo Fukushima, and Takeshi Naemura. 2017. Imperceptible Color Vibration for Embedding Pixel-by-Pixel Data into LCD Images. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI EA '17). Association for Computing Machinery, New York, NY, USA, 1464–1470. doi:10.1145/3027063.3053074
- [2] Satoshi Abe, Takefumi Hiraki, Shogo Fukushima, and Takeshi Naemura. 2018. Screen-Camera Communication via Matrix Barcode Utilizing Imperceptible Color Vibration. In *Adjunct Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (Berlin, Germany) (UIST '18 Adjunct). Association for Computing Machinery, New York, NY, USA, 166–168. doi:10.1145/3266037.3271638
- [3] Hideyuki Ando, Junji Watanabe, Tomohiro Amemiya, and Taro Maeda. 2007. Full-Scale Saccade-Based Display: Public / Private Image Presentation Based on Gaze-Contingent Visual Illusion. In *ACM SIGGRAPH 2007 Emerging Technologies* (San Diego, California) (SIGGRAPH '07). Association for Computing Machinery, New York, NY, USA, 5–es. doi:10.1145/1278280.1278286
- [4] Oliver Bimber and Ramesh Raskar. 2005. *Spatial Augmented Reality: Merging Real and Virtual Worlds*. A. K. Peters, Ltd., USA.
- [5] Haruka Ikeda, Tomohiko Hayakawa, and Masatoshi Ishikawa. 2019. Bilateral Motion Display: Strategy to Provide Multiple Visual Perception Using Afterimage Effects for Specific Motion. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology* (Parramatta, NSW, Australia) (VRST '19). Association for Computing Machinery, New York, NY, USA, Article 17, 5 pages. doi:10.1145/3359996.3364241
- [6] Mariko Isogawa, Daisuke Iwai, and Kosuke Sato. 2014. Making Graphical Information Visible in Real Shadows on Interactive Tabletops. *IEEE Transactions on Visualization and Computer Graphics* 20, 9 (2014), 1293–1302. doi:10.1109/TVCG.2014.2316002
- [7] Arisa Kohtani, Shio Miyafuji, and Hideki Koike. 2025. Color Display on Moving Object Trajectories Using High-Speed (IEEEVR '25). IEEE.
- [8] Arisa Kohtani, Shio Miyafuji, Keishiro Urakaki, Hidetaka Katsuyama, and Hideki Koike. 2024. MOSion: Gaze Guidance with Motion-triggered Visual Cues by Mosaic Patterns. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 374, 11 pages. doi:10.1145/3613904.3642577
- [9] Natalia D. Mankowska, Anna B. Marcinkowska, Monika Waskow, Rita I. Sharma, Jacek Kot, and Pawel J. Winkowski. 2021. Critical Flicker Fusion Frequency: A Narrative Review. *Medicina* 57, 10 (2021). doi:10.3390/medicina57101096
- [10] Fumihiko Sakaue and Jun Sato. 2020. Active 3D Motion Visualization Based on Spatiotemporal Light-Ray Integration. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1977–1985. doi:10.1109/CVPR42600.2020.00205
- [11] Junji Watanabe, Taro Maeda, and Hideyuki Ando. 2012. Gaze-Contingent Visual Presentation Technique with Electro-Ocular-Graph-Based Saccade Detection. *ACM Trans. Appl. Percept.* 9, 2, Article 6 (jun 2012), 12 pages. doi:10.1145/2207216.2207217
- [12] Hirotsugu Yamamoto, Kengo Sato, Syahmi Farhan, and Shiro Suyama. 2014. 62.3: Hand-Waving Steganography by Use of a High-Frame-Rate LED Panel. *SID Symposium Digest of Technical Papers* 45, 1 (2014), 915–917. doi:10.1002/j.2168-0159.2014.tb00241.x arXiv:https://sid.onlinelibrary.wiley.com/doi/pdf/10.1002/j.2168-0159.2014.tb00241.x
- [13] Aya Yohso and Kazuhiko Ukai. 2006. How color break-up occurs in the human-visual system: The mechanism of the color break-up phenomenon. *Journal of The Society for Information Display - J SOC INF DISP* 14 (12 2006). doi:10.1889/1.2408396